ModelIt!

Student Activity Pack

Earth's Hidden Water

97% Is Undrinkable, 2% Is Frozen. We Live on the Rest.

Grade 5 | NGSS 5-ESS2-2 | Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Shade a pie chart to show what percent of Earth's water is saltwater, frozen, and drinkable.**

2. **What controls whether a city has enough water to drink?**

3. **Have you heard of a city running out of water? What happened?**

4. **Why do you think some places have water shortages while others don't?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Precipitation Amount | Temperature | Freshwater Availability | Evaporation Rate

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Normal Balance

Moderate precipitation and temperature

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Drought + Heat Wave

Precipitation LOW, Temperature HIGH

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Winter Snowpack

Precipitation HIGH, Temperature LOW

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Human Water Use — Agriculture uses 70% of all fresh water

Groundwater Pumping — Underground water being used faster than it refills

Glacier Melt — Ice melting provides temporary water before disappearing

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What controls how much fresh water is available for communities to use? Use evidence from your simulation to support your answers.

1 Why does California depend so heavily on mountain snowpack?

2 What happens to freshwater availability as global temperatures rise?

3 Which conditions create the greatest risk for communities?

4 How is the water you drink connected to weather hundreds of miles away?

5 What is one thing humans do that decreases freshwater availability?

8 STEM Challenge: Design a Water Conservation Plan

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YOUR ENGINEERING MISSION

Design a water conservation plan for a city facing drought conditions.

THE SCENARIO

A city is running low on water. They need to reduce usage by 40% without hurting residents.

Engineering Design Thinking

1. WHAT ARE THE BIGGEST WATER USES IN A CITY?
2. WHICH CUTS WOULD HAVE THE LEAST IMPACT ON DAILY LIFE?
3. HOW WOULD YOU COMMUNICATE RESTRICTIONS TO RESIDENTS?
4. WHAT INCENTIVES COULD ENCOURAGE CONSERVATION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Hydrologists and Water Resource Managers protect community water supplies and predict droughts. They earn $75,000-$110,000/year.